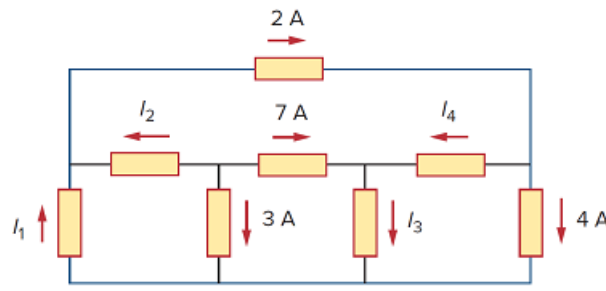


Problem

For the circuit in Fig. 2.77, use KCL to find the branch currents I_1 to I_4 .

Figure 2.77:



Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

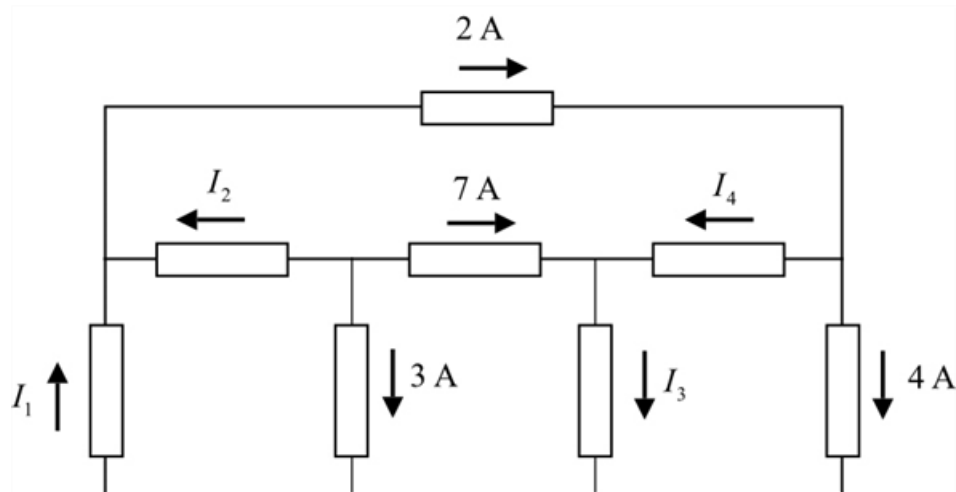


Figure 1

Step 2 of 6

Redraw the circuit with node notations:

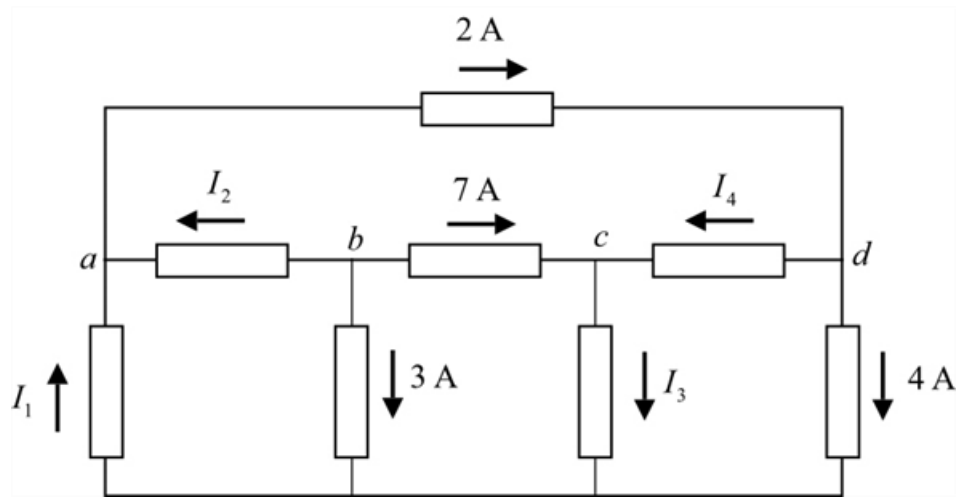


Figure 2

Step 3 of 6

Apply Kirchhoff's current law at node d .

$$2 - I_4 - 4 = 0$$

$$I_4 = -2 \text{ A}$$

Therefore the current I_4 is .

Step 4 of 6

Apply Kirchhoff's current law at node c .

$$I_4 - I_3 + 7 = 0$$

Substitute 2 for I_4 in the above equation.

$$-2 - I_3 + 7 = 0$$

$$I_3 = 5 \text{ A}$$

Therefore the current I_3 is .

Step 5 of 6

Apply Kirchhoff's current law at node b .

$$-3 - I_2 - 7 = 0$$

$$I_2 = -10 \text{ A}$$

Therefore the current I_2 is .

Step 6 of 6

Apply Kirchhoff's current law at node a .

$$I_1 + I_2 - 2 = 0$$

Substitute 10 for I_2 in the above equation.

$$I_1 - 10 - 2 = 0$$

$$I_1 = 12 \text{ A}$$

Therefore the current I_1 is 12 A.